

AI Communication and Confidence Analysis Engine

Overview

The AI Communication and Confidence Analysis Engine extends the Zecpath AI interview system by evaluating how candidates communicate during interview interactions.

The implementation focuses on analyzing response quality, confidence-related signals, sentiment, fluency, vocabulary, clarity, filler words, grammar indicators, response structure, consistency, and speaking pace when duration information is available.

The system is designed as a supporting evaluation layer for AI-driven interviews and does not replace human decision-making.

Confidence and Sentiment Analysis

The ConfidenceSentimentEngine analyzes candidate responses using configurable signals defined in the confidence and sentiment configuration.

The evaluation includes:

- Hesitation detection
- Uncertainty detection
- Response length analysis
- Speaking pace analysis
- Positive sentiment detection
- Negative sentiment detection
- Consistency comparison with previous answers
- Confidence score calculation
- Communication strength classification

The confidence score is maintained within a range of 0 to 100.

Communication strength is classified into:

- Strong
 - Moderate
 - Needs Improvement
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Communication Scoring

The CommunicationScoringEngine evaluates the overall quality of a candidate's communication.

The engine produces a communication score and communication level based on multiple metrics.

The evaluated metrics include:

Fluency

Fluency analysis considers sentence count and average sentence length to estimate how naturally the response is structured.

Grammar

The grammar metric identifies basic grammar-related issues and produces a grammar score.

Vocabulary

Vocabulary analysis considers word count, unique words, and vocabulary diversity.

Clarity

The clarity metric evaluates whether the candidate response is sufficiently understandable and appropriately expressed.

Fillers

The system detects filler expressions that may reduce communication quality.

Structure

Response structure considers sentence organization and whether the response contains an understandable flow.

Structured Output

The CommunicationScoringEngine produces structured output containing:

- Raw response text
 - Normalized response text
 - Communication score
 - Communication level
 - Fluency metrics
 - Grammar metrics
 - Vocabulary metrics
 - Clarity metrics
 - Filler metrics
 - Structure metrics
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Integration with Interview AI

The confidence and communication engines complement the existing interview workflow.

The broader workflow includes:

1. Role-based interview question generation
2. Interview conversation phases
3. Candidate response capture

4. Dynamic follow-up generation
5. Confidence and sentiment analysis
6. Communication quality scoring

Together, these components provide a stronger foundation for adaptive AI-assisted interview evaluation.

Testing

Dedicated automated tests were created for the CommunicationScoringEngine.

The tests verify:

- Engine initialization
- Basic response analysis
- Communication score range
- Fluency metrics
- Grammar metrics
- Vocabulary metrics
- Clarity metrics
- Filler detection
- Structure metrics
- Communication level classification
- Empty response handling
- Structured output

The communication scoring test suite completed successfully with 12 passing tests.

A full regression test was also executed across the confidence, communication, interview, and follow-up components.

The complete regression suite completed successfully with 48 passing tests.

Result

The Zecpath AI interview system now includes communication and confidence evaluation capabilities alongside its existing interview engine and adaptive follow-up functionality.

This creates a more comprehensive interview analysis pipeline and provides a foundation for future integration of candidate performance scoring and interview decision support.